Fake News Detection Analysis

Exploratory Data Analysis Report

Executive Summary

This report compares about 39,000 fake and true news articles to find signals that could help detect fake news. The clearest difference is in how the articles are written: true articles rely much more on named sources (the word "said" is far more prominent in them), while fake articles lean on a more informal, opinion-style vocabulary and are slightly longer on average. A practical detection approach should be built on writing style and sourcing, not on source labels.

Objective

To understand how fake news articles differ from real ones, and to identify signals that a fake news detection approach could use, while spotting any data issues that would make a detector look accurate for the wrong reasons.

Data Description

Dataset	Articles	Columns
Fake.csv	23,481	title, text, subject, date
True.csv	21,417	title, text, subject, date

The two files were combined, with a label column added to record whether each article is fake or true. The analysis was done in Python using pandas, matplotlib, seaborn and NLTK.

Data Cleaning and Preprocessing

- **Duplicates:** 5,793 exact duplicate articles (same title and text) were removed, leaving 39,105 articles. Most of the duplicates came from the fake file.
- **Class balance:** before cleaning, the data was 52.3% fake and 47.7% true. Removing the duplicates leaves slightly more true articles than fake, but the classes remain reasonably balanced, so no resampling is needed.
- **Near-empty articles:** 474 articles have almost no body text (under 20 characters), and almost all of them are fake. They were flagged and kept, because removing them would quietly shrink the fake class.
- **Dates:** dates were stored in different formats and some had trailing spaces, so they were cleaned and standardised. Six rows had a web address or headline in the date field instead of a date and were left as missing.
- **Text cleaning:** a separate cleaned copy of the text was created (lowercase, web addresses, punctuation and numbers removed) for word analysis, so the original text was kept untouched. Common English stopwords such as "the" and "and" were excluded from word counts.

Key Findings

Source labels separate the classes perfectly

Subject	Fake	True
News	9,050	0
politics	6,838	0
US_News	783	0
left-news	705	0
Government News	532	0
politicsNews	0	11,217
worldnews	0	9,980

No subject appears in both classes, so subject only tells us which source file an article came from. The word "reuters" is also among the most common words in the true articles (see the top words chart in the notebook), which fits all the true articles coming from one news agency. The two classes also cover different periods: fake articles run from 2015 to early 2018, while true articles only run from January 2016 to December 2017.

Sourcing and article length

Measure	Fake	True
Average article length (words)	422	387
Median article length (words)	378	360

- **Sourcing:** the word "said" is far more prominent among the most common words in true articles than in fake ones, which fits wire-style reporting where claims are tied to named people or organisations.
- **Article length:** fake articles are slightly longer on average, but the difference is small, so length is a very weak signal.

Vocabulary

Both classes are dominated by political words such as "trump" and "president". Fake articles lean towards informal and opinion words (people, one, like, even, would), while true articles lean towards institutional words (government, state, house, republican, states).

Articles over time

Fake articles grow from about 160 a month in 2015 to roughly 550 to 900 a month from 2016 to mid-2017, then decline to about 300 by December 2017. True articles run at about 250 to 750 a month until August 2017, then jump to about 3,000 a month from September to November 2017.

Insights

1. **Sourcing language is the strongest content signal.** True articles use attribution words, especially "said", far more than fake ones. Real reporting ties claims to named sources, and this is easy to explain to a reader.
2. **Writing style differs.** Fake articles use more informal, opinion-style language, while true articles use more institutional vocabulary.
3. **Article length is not a reliable signal.** The difference between classes is small, and fake articles are slightly longer.
4. **Near-empty articles are almost all fake.** Of 474 articles with almost no body text, almost all are fake, so a title with no real body is a warning sign.
5. **Subject and date would make a detector look accurate for the wrong reasons.** They identify which source file an article came from (the two classes were collected from different sources and over different periods), not whether it is fake. A model using them would score highly on this data and then fail on news from other outlets.

Recommendations

1. **Build the detector on content and style, not on metadata.** Use features such as how often claims are attributed to a source and the vocabulary of the article, and test whether sensational or emotive language in the title adds further signal.
2. **Remove leakage features before any modelling.** Exclude subject and date from the features, and remove source names such as the news agency's name from the text, so the model has to learn from content.
3. **Flag suspicious articles for human review.** Articles with little or no body text, especially with a sensational title, should be queued for manual checking, not published or trusted automatically.
4. **Use an explainable risk score.** Combine sourcing and vocabulary signals into a simple score and show editors the reasons behind it, because readers and moderators are more likely to trust a decision they can understand.
5. **Test on new data.** Validate any detector on articles from other outlets and other time periods, because here all true articles come from one outlet and both classes come from a limited period.
6. **Next step: build a simple baseline model.** After removing the leakage features, train a baseline classifier on the cleaned text and measure precision and recall, so the signals found here can be tested rather than assumed.

Limitations

- All true articles come from a single news agency (Reuters), so the dataset cannot show how fake news differs from real news in general.
- The data covers political and world news from 2015 to 2018 and may not represent other topics or later years.

Conclusion

Fake and true articles in this dataset differ most in sourcing and writing style, and only slightly in length. Real articles attribute their claims, while fake articles lean on informal, opinion-style language and are slightly longer. The data also contains columns (subject and date) that identify

the source file rather than the truth of an article, so a trustworthy detector must be built on writing style and sourcing, and tested on news from other outlets. The recommendations above give a practical starting point for a simple, explainable detection approach.